

How much of the evidence for climate-driven dengue transmission survives the analyst?

A multiverse analysis of 221 outbreaks in 33 countries

Amna Dilber

Independent research

amnadilber.bi@gmail.com

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Abstract

Compartmental models fitted to surveillance data are routinely used to test whether dengue transmission is climate-driven, and reviews explain their conflicting findings by appealing to local context. We fit every usable outbreak in a global surveillance compilation under all 144 combinations of six analysis choices that published models rarely state, so that no choice varies with the setting because none varies at all: 33,152 fits, of which the 221 outbreaks in 33 countries completing every combination are analysed.

Roughly three-quarters of the variation in whether climate forcing is endorsed lies between analyses of the same outbreak, not between outbreaks (76.5%; 95% CI 72.6–81.1 resampling whole countries). Partitioned three ways the largest term is neither: **61%** is an outbreak-by-analysis interaction, against 16% for the analysis itself, so no standardised convention can reach it and a specification curve, which displays the main effect, cannot show it. Adding climate covariates moves estimated R_0 the same way, its interaction share rising from 23% to 69%.

The intervals do not keep their promise. A nominal 95% interval contains the true transmission coefficient **78%** of the time and the true rainfall coefficient **71%** — **42%** when the epidemic is a sum of two asynchronous ones, as these data demonstrably are. Combining **eight** analyses by Rubin’s rules restores nominal coverage at 1.5–2.2 times the width, reaching 90% rather than 95% under structural mis-specification.

Applying our own rule gives a clear verdict for 137 of 221 outbreaks, **86% of which support climate forcing**, and the three-way partition of the real data matches simulated outbreaks containing a real effect rather than simulated outbreaks containing none. The claim is not that the effect is absent, but that in 38% of outbreaks a single season cannot settle the question.

1 Introduction

Whether dengue transmission is driven by climate is asked repeatedly of surveillance data, and the answer feeds early-warning systems, attribution studies and control policy. The usual method is to fit a transmission model with and without climate covariates and compare them by an information criterion.

Such an analysis requires decisions that the model itself does not determine. A distribution must be chosen for the observation error. A functional form must be chosen for the temperature response. Rainfall must be lagged by some number of weeks — a lag with a mechanistic basis and no agreed value, since both drought and extreme rainfall raise dengue risk at different delays [16]. Some portion of the series must be set aside for validation, or none. None of these is a hypothesis about dengue; each is a convention, and published papers rarely state why one was preferred over another.

A systematic review of 99 dengue prediction models from 64 studies found that “[t]he reporting of methodology and model performance measures were inadequate in many of the existing prediction models” [1]. The field is aware of the problem. What has not been quantified is its consequence: how often the reported conclusion would have differed had the analyst chosen otherwise.

We measure this directly. Every usable dengue outbreak in a global surveillance compilation — 221 of them, in 33 countries — is fitted under a full factorial of six such choices, 144 combinations and 33,152 fits, so that no analytical choice varies with the setting because none varies at all. Four things follow.

1. **The disagreement in this literature is mostly not geographic.** Roughly three-quarters of the variation in whether an outbreak is judged climate-driven lies between analyses of the same outbreak rather than between outbreaks (Section 4.1). Everything that differs between epidemics competes for the remaining quarter, which bounds what any contextual explanation can achieve — and the specific contextual explanation on offer does not reproduce here. Partitioning three ways, the largest term is an outbreak-by-analysis interaction at 61%, four times the main effect of the analysis: the disagreement is not a ranking of methods that a house style could settle.
2. **The cost falls on the number the field reports.** Under a constant model, 72% of the variation in estimated R_0 lies between outbreaks, which is what an estimate is for. Adding climate covariates inverts that: 80% or more of the variation is then between analyses of the same outbreak (Section 4.2).
3. **The intervals do not keep their promise.** A nominal 95% interval contains the true transmission coefficient 78% of the time, and the true rainfall coefficient 71% — 42% when the epidemic is a sum of two asynchronous ones, which these data demonstrably are (Section 4.12).
4. **An interval that does keep it, validated against a known truth and cheap enough to use.** Combining the analyses by Rubin’s rules restores nominal coverage at 1.5 to 2.2 times the width, and **eight analyses are enough** — four already reach nominal. It is an improvement rather than a cure: under structural mis-specification it falls to 90%, and we report that rather than the headline alone.

The diagnosis is therefore accompanied by an instrument, and the instrument has been tested where it fails as well as where it works.

Relation to multiverse and specification-curve analysis. The design is a multiverse analysis in the sense of Steegen et al. [6] and a specification curve in the sense of Simonsohn et al. [7]: enumerate the defensible analyses rather than report one, and treat the spread of results as the quantity of interest. The complementary design hands one dataset to many analysts and observes what they do with it — 29 teams reaching effect sizes from 0.89 to 2.93 on one question [10], and 70 teams producing sizeable variation in hypothesis tests from one neuroimaging dataset [11]. Both literatures have reached epidemiology: a specification curve over 1,208 analyses of red meat and mortality found 36% of them above the null and 64% below [12]. Those literatures developed around regression models, where the forking paths are covariate sets, exclusion rules and outcome codings. Mechanistic transmission models fork differently. The choices here are the observation distribution, the functional form of a biological response, a lag with a mechanistic justification but no agreed value, the compartmental structure, and the values at which the epidemiological parameters are fixed — none of which has an analogue in a covariate list, and each of which is normally presented as part of the model rather than as an analytical degree of freedom. Two consequences follow that a regression multiverse does not

face: the analyses can fail to converge, so the multiverse is not guaranteed to be complete; and the comparison is between nested mechanistic hypotheses judged by an information criterion, which makes the decision rule itself — where the threshold sits, and whether it abstains — a lever we can test rather than assume. To our knowledge this framework has not previously been applied to mechanistic epidemic models.

The design also lets us report something a specification curve cannot. A curve orders specifications by their average result across datasets, which is a main effect of the analysis; with many datasets analysed every way, the dataset-by-specification interaction is separately estimable, and here it is the largest of the three terms (Section efsec:decomposition). Variance decompositions over analytic *factors* appear in the multiverse literature; we have not found the interaction with the dataset reported, though the multi-pipeline neuroimaging collections have the crossed design in which it could be. It matters because it decides which remedy is available: a main effect can be legislated away by convention and an interaction cannot.

2 Data and outbreak selection

Weekly reported dengue case counts were taken from OpenDengue v1.3 [2], a curated compilation of publicly available surveillance data covering 88 countries in its weekly-resolution subset (2,248,637 records). Daily temperature and precipitation were taken from NASA POWER at one representative location per reporting unit — the capital or largest city for a national series, the administrative seat for a subnational one.

A window is usable only if it can carry a single-epidemic fit. Each reporting unit’s longest unbroken run of weekly observations was split into waves outward from each peak, extending until counts fell below a quarter of that peak, with eight quiet weeks retained on either side to anchor the initial condition and the final size. A window was kept if it spanned 30–110 weeks, contained at least 300 cases, peaked at least 20 cases in a week and at least $2.5\times$ the median week, with the peak interior to the window.

The peak-prominence threshold is itself a choice, and Section 4.14 reports the headline result across a range of values rather than only the one applied. The procedure yields **237 windows from 34 countries**, median 38 weeks and 4,318 cases: 125 national and 112 subnational. Four Brazilian windows could not be fitted under any configuration — every multi-start diverged — and twelve further windows completed only part of the factorial. Both groups are excluded, leaving **221 outbreaks in 33 countries** and 33,152 fits. A window finishing part of the design would appear stable or unstable depending on which cells happened to fail rather than on the choices.

Requiring an unbroken weekly run is restrictive, and Section 5 discusses what it excludes.

3 Model and factorial design

Dengue is vector-borne, so a single-population model omits the delay imposed by the mosquito’s own infection cycle. A systematic review of dengue transmission models catalogues the structural choices this implies and finds no settled convention among them [15]. We use a host–vector model: humans move through susceptible, exposed, infectious and recovered compartments, mosquitoes through susceptible and infectious without recovery. With state variables as fractions of their own population,

$$\dot{S}_h = -\beta(t) m I_v S_h, \quad \dot{S}_v = \mu_v(1 - S_v) - \beta(t) I_h S_v, \quad \dot{E}_h = \beta(t) m I_v S_h - \sigma_h E_h, \quad \dot{I}_v = \beta(t) I_h S_v - \mu_v I_v, \quad \dot{I}_h = \sigma_h E_h - \gamma_h I_h$$

where m is the vector-to-host ratio, μ_v the mosquito mortality rate, $1/\sigma_h$ the intrinsic incubation period and $1/\gamma_h$ the infectious period. Mosquito births replace deaths so the vector population is constant; seasonal abundance is carried by $\beta(t)$ rather than by a second time-varying

state, which keeps the state space small and avoids a poorly identified quantity. Cumulative incidence C is what surveillance detects, up to the reporting fraction. Expected weekly counts are $\rho N [C(t+7) - C(t)]$.

The climate model sets

$$\beta(t) = \beta_0 B(T(t))^{a_{\text{temp}}} \exp(a_{\text{rain}} z(t)),$$

with B the Brière unimodal thermal response [13] normalised to one at its peak, T weekly mean temperature and z standardised lagged rainfall; the constant model sets $\beta(t) = \beta_0$. Writing temperature as an exponent rather than a free coefficient makes it a nested hypothesis: at $a_{\text{temp}} = 0$ temperature drops out, at 1 the literature response applies in full. The alternative log-linear parameterisation replaces $B(T)^{a_{\text{temp}}}$ with $\exp(a_{\text{temp}} z_T)$ on standardised temperature.

The human-only SEIR structure removes the vector, giving $\dot{S}_h = -\beta(t) I_h S_h$ and the remaining human equations unchanged. The third structure is the host–vector system above with μ_v replaced by $\mu_v / \max(L(T(t)), 0.15)$, where L is a quadratic lifespan response normalised to one at its peak, so that mortality equals its constant value at the thermal optimum and rises on either side. Mosquito lifespan and the incubation periods are fixed from the entomological literature, since human case counts cannot identify them.

Two models are compared in every window. The *climate* model makes the composite transmission coefficient depend on temperature and lagged rainfall; the *constant* model holds it fixed. They are nested, and the constant model is the honest baseline against which any claimed climate effect must be judged.

One structural point constrains what can be estimated at all. The state variables are fractions, so the population enters predictions only where incidence is converted to counts, and the reporting fraction ρ and the population at risk N appear solely as the product ρN . No data can separate them; we verified numerically that halving N while doubling ρ changes every predicted week by exactly zero. We therefore fix ρ from the under-ascertainment literature — cartographic estimates put global dengue infections at about three times the apparent cases and four times the WHO figure [17] — and estimate the population at risk. Varying ρ over a tenfold range rescales that estimate as $1/\rho$ and leaves R_0 unchanged to four decimal places, so no conclusion here depends on it.

Estimation. The system is integrated with fixed-step classical Runge–Kutta at one day, whose adequacy under climate forcing is verified against an eightfold finer step. Five parameters are estimated — β_0 , a_{temp} , a_{rain} , the fraction of the population at risk and the initial infected fraction — by maximum likelihood, expressed as least squares on signed square roots of the per-observation deviance so that a Levenberg–Marquardt routine performs Poisson or negative-binomial estimation according to which residual is supplied. Optimisation runs in an unconstrained space, log for positive parameters and logit for fractions, from three starts; the negative binomial’s dispersion is profiled. A state leaving $[0, 1]$ raises an integration failure that the optimiser converts to a flat penalty rather than a gradient, so a divergent proposal cannot be followed downhill.

Standard errors are asymptotic, from the Jacobian at the optimum, inflated by the Pearson dispersion and transformed back to the natural scale by the delta method. These are the intervals whose coverage Section 4.12 measures: they are the ones a paper of this kind reports.

The factorial. Each window is fitted under all combinations of:

- **observation model** — Poisson, or negative binomial with the dispersion profiled;
- **temperature term** — a unimodal Brière response with thermal limits fixed from *Aedes aegypti* trait data [4, 14], raised to an estimated exponent; or a free log-linear coefficient on standardised temperature, the conventional choice;

- **rainfall lag** — 3, 5 or 7 weeks;
- **fraction of the series fitted** — 100%, or the first 75% with the remainder held out;
- **model structure** — three formulations. The host–vector model above; a directly transmitted human-only SEIR, which is what a large part of the applied literature fits to case counts; and the host–vector model with adult mosquito mortality following temperature. The third exists because holding mosquito lifespan constant across a season whose temperature swings by fifteen degrees is not what mechanistic dengue modelling does, and a robustness result obtained only on models simpler than current practice would say little about current practice. Its thermal limits are fixed from vector biology and it estimates no additional parameter, so the factor remains a comparison between mechanisms rather than between one model given more freedom than another;
- **fixed epidemiological parameters** — two sets, both drawn from published ranges: human viraemia 4 or 5 days, intrinsic incubation 5 or 6 days, mosquito lifespan 10 or 14 days, vector-to-host ratio 2 or 3. Every mechanistic model fixes these from the literature, every published range is wide, and choosing within one is a degree of freedom exactly like the rainfall lag. Earlier versions of this study fixed them and listed the choice as a limitation, which would have made its own headline a lower bound for a reason it could remove.

Structures are fitted independently, each estimating its own transmission coefficient; they are not constrained to a common R_0 . The sampling range for that coefficient is shared, implying R_0 ranges of roughly $[0.01, 20]$ for the host–vector structures and $[0.005, 10]$ for SEIR, both wide enough to contain any plausible dengue value, so the shared range does not bind one structure more than another.

What the null model means in the third structure. There, temperature enters *both* compared models, since it drives mosquito mortality in the structure itself. Its constant model is not climate-free: it is “climate affects mosquito survival but not the transmission coefficient”. The comparison is the same in all three — does letting the transmission coefficient depend on climate improve the fit, holding the mechanism fixed — but the baseline is richer in the third, which makes it the hardest of the three tests rather than an unfair one. We state this because a structure whose null is not the same null would otherwise read as a like-for-like comparison.

One hundred and forty-four combinations, two models each. Windows that did not complete every combination are excluded: a window finishing part of the design would appear stable or unstable depending on which cells happened to fail rather than on the choices.

Are these the choices the literature actually makes? Everything in this paper rests on the factor list being a description of practice rather than of our imagination, so the evidence for it should be given rather than asserted. The systematic review of 99 dengue prediction models cited above [1] reports its own breakdown, and two of our factors appear in it directly. Among the statistical models, **18.3% used Poisson regression and 18.3% linear regression**; the negative binomial does not appear as a category of its own. A linear model on counts assumes constant variance and a Poisson model assumes variance equal to the mean, so more than a third of that literature is fitted under an assumption our simulations show endorses climate forcing on data containing none. Mechanistic fits, which the review does not cover, commonly go further and use ordinary least squares.

The same review reports that **20.2% of models present no validation at all**, 75.8% internal validation only and 5.2% external — so how much of a series to fit is not a settled convention either.

For the remaining four factors the evidence is the specifications themselves: every mechanistic model must choose a functional form for the temperature response, a rainfall lag, a compartmental structure and values for the fixed entomological parameters, and every published range for those parameters is wide. What we cannot show is that our two or three settings per factor span what the literature does; they do not, and Section 5 says so.

4 Results

4.1 Three-quarters of the disagreement is not about the place

Reviews of this literature explain its conflicting findings by appealing to local context: one setting is more variable, another denser, a third poorer. A recent meta-analysis of 358 published temperature–dengue correlations formalised that explanation, finding the association strongest where temperature varies most [5]. The explanation requires the variation between studies to sit between *settings*. Whether it does has not been checkable, because each published estimate comes from one analysis of one place and the two cannot be separated after the fact.

Here they can. Every outbreak has been analysed every way, so the variation in whether climate forcing is endorsed splits exactly:

Table 1: Where the disagreement lives. “Between” is the variance of the per-outbreak means; “within” the mean of the per-outbreak variances. Unlike the share of outbreaks in which some pair disagrees, this split does not inflate as the factorial grows.

Design	Analyses	Between outbreaks	Within outbreaks
Four factors	24	21.7%	78.3%
Five factors	48	20.3%	79.7%
Six factors	144	23.5%	76.5%

Roughly three-quarters of the variation is between analyses of the same outbreak.

Everything that differs between epidemics — climate, population density, wealth, serotype composition, health system, surveillance intensity — competes for the remaining quarter, and that quarter is an upper bound on what any moderator analysis can hope to explain. The split moves by three points across designs of 24, 48 and 144 analyses, so it is not an artefact of how finely the space was enumerated.

Bootstrapping over outbreaks gives a 95% interval of 73.1–79.9%. The outbreaks are not independent — 221 of them come from 85 reporting units in 33 countries, and three countries supply 45% of the sample — so the interval was recomputed resampling whole countries rather than outbreaks. It widens to 72.6–81.1% and the point estimate does not move.

Is this a fact about Latin America? Three-fifths of the sample is, which is a property of where weekly dengue surveillance is published rather than a choice we made, and it has to be checked rather than excused. Removing each well-represented country in turn moves the share between 76.0% and 78.6%. Removing all three dominant countries at once — Bolivia, Nicaragua and Mexico, leaving 121 outbreaks — gives **78.5%**. Splitting by reporting level gives 79.1% for national series and 75.4% for subnational. Every subset we can form lands within three points of the whole.

How much does this depend on which choices we called defensible? More than the table above admits, and the honest version is worth more than the comfortable one. The three designs are nested — all contain the observation model — so agreement between them shows the answer survives adding cells to one factor set, not that it survives choosing another. We

therefore recomputed it dropping each factor in turn, and over all 63 non-empty subsets of the six.

Table 2: The within-outbreak share under alternative factor lists. The result is load-bearing on the observation model and does not survive an adversarial choice of factor list, which is stated here rather than left to be found.

Factor list	Within-outbreak share
All six factors	76.5%
Dropping the fixed parameters	76.7%
Dropping the model structure	73.6%
Dropping the rainfall lag	73.0%
Dropping the temperature form	71.4%
Dropping the fraction of series fitted	70.5%
Dropping the observation model	55.2%
Median over all 63 subsets	54.5%
Best case for a sceptic (fixed parameters alone)	2.3%

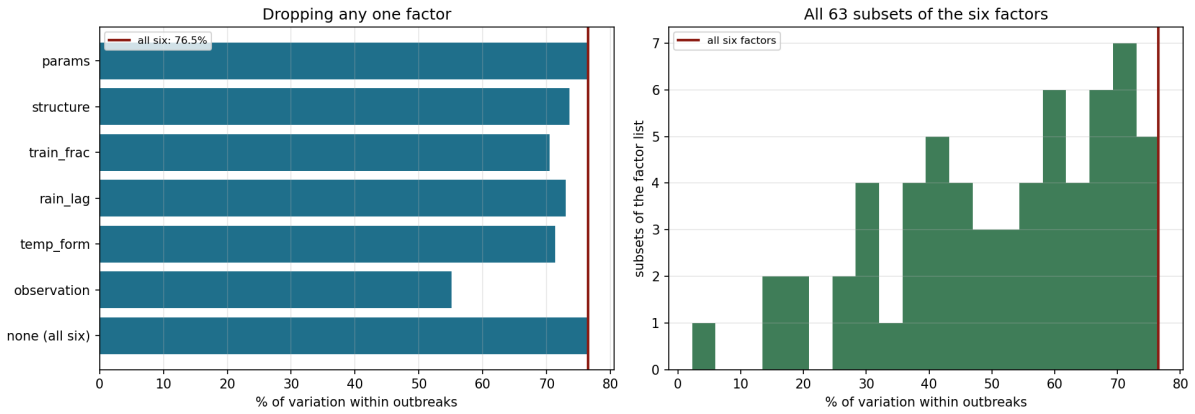


Figure 1: How far the headline depends on the factor list. Left: the within-outbreak share when each factor is dropped in turn. Right: the same quantity over all 63 non-empty subsets of the six factors. The claim is load-bearing on the observation model and does not survive an adversarial choice of factor list, which is why both panels are shown rather than the first alone.

Removing any factor except the observation model costs at most six points. Removing the observation model costs twenty-one, and the analytical share is then 55.2% — still a majority, but the claim is plainly load-bearing on that one choice. A reader who regards the observation model as fixed by good practice rather than chosen should read 55%, and one who regards only the entomological parameters as genuinely free should read 2.3%.

We think the first two positions are the defensible ones, and the evidence for that is in Section 3: the published review finds 18.3% of models using Poisson regression and 18.3% linear regression, with the negative binomial not appearing as a category. The observation model is not fixed by practice, because practice has not fixed it.

One feature of the design was said to push the other way, and the claim was never tested. Every combination within an outbreak is warm-started from a single anchor fit and shares that fit’s estimate of the dispersion, so the 144 analyses are arguably more alike than 144 independent analysts would be, and earlier drafts concluded that the share reported here is if anything an underestimate. The optimiser check of Section 4.14 refits seven outbreaks from ten cold starts

with no warm start, which measures the direction rather than arguing it: pairwise disagreement is **33.4%** as run and **32.9%** cold. Cold starts give slightly *less* disagreement, not more. Seven outbreaks settle nothing about the size, and the check varies the warm start but not the shared dispersion, so one of the two mechanisms remains untested. We no longer claim to know the sign.

This does not show that local context is irrelevant. It shows that context has been asked to carry more than it can.

Neither the place nor the method: the term that is both. The split above has a remainder, and the remainder is not the analysis. “Between” is the variance of the per-outbreak means; “within” is everything else, and everything else contains two different things. A *main effect* of the analysis would mean that some conventions are systematically more credulous than others, which stating and standardising the choices would remove. An *interaction* would mean that which analysis endorses depends on which outbreak, which no convention can remove. The design is completely crossed and complete by construction, so the sum of squares partitions exactly into the three.

Table 3: Where the variation in the verdict lives, partitioned three ways. The shares sum to one exactly. Intervals resample whole countries.

Design	Outbreak	Analysis	Interaction
Four factors, 24 analyses	21.7%	19.6%	58.7%
Five factors, 48 analyses	20.3%	21.2%	58.5%
Six factors, 144 analyses	23.5%	15.9%	60.6%
95% CI	18.6–27.7	11.9–22.9	56.1–63.8

The largest term is the one that belongs to neither. Three-fifths of the variation in whether an outbreak is judged climate-driven is specific to the pairing of a method with an epidemic: not a ranking of methods, not a property of places. It is the most stable of the three across designs — 58.7%, 58.5% and 60.6% on 24, 48 and 144 analyses — and removing each dominant country in turn moves it between 59.7% and 62.1%.

An interaction estimated with one observation per cell cannot be separated from error by the design, so it is worth saying what the error here is. Each cell is a single deterministic fit and the only noise available is the optimiser’s; the check in Section 4.14 changes 0.6% of verdicts under ten cold restarts, which bounds that contribution at about one part in a hundred of a term worth three-fifths.

This is also what a specification curve cannot show. The curve orders specifications by their average result across datasets, which is the definition of a main effect, and averages the interaction away. The 56-percentage-point spread reported in Section 4.5 is real, and it is the smaller of the two findings: the term on display is a quarter of the term it hides.

Put as two published findings would meet each other: two analyses of the same outbreak reach opposite verdicts 29.8% of the time, the *same* analysis applied to two different outbreaks 32.7%, and two findings sharing neither 38.9%. Standardising the analysis removes 6 points of that disagreement; standardising the place removes 9. An earlier version of this paper claimed that a conflicting pair of published findings is more likely to differ because the analysts differed than because the places did. We withdraw it. The one-way split does not license it — a variance of means and a mean of variances are not comparable in that direction — and measured directly it is if anything the wrong way round.

Is the interaction predictable, or only unexplained? An interaction is a negative result until someone asks whether it can be anticipated, and the two answers point opposite ways. If

an analyst could tell in advance which choices will matter for their data, the interaction is a tool. If not, there is nothing to report but the spread.

One prediction has a direction fixed before it was tested. The observation model is the dominant factor and the reason a Poisson likelihood misbehaves is that it denies overdispersion, so its effect should be largest where the counts are most overdispersed. Dispersion is measured here by the index of Section 4.8, taken about a five-week moving average, which depends on no transmission model and enters no cell of the factorial.

The prediction holds. Across the 221 outbreaks the effect of switching to Poisson correlates with log dispersion at $\rho = +0.31$ ($p = 2 \times 10^{-6}$): below the median it raises the endorsement rate by **0.25** and above it by **0.42**. The interaction is not arbitrary where a mechanism exists — it is largest exactly where the assumption it rests on is most violated. That is independent corroboration of the mechanism in Section 4.4 from a measure that touches none of the same arithmetic.

And it stops there. Regressing each factor’s outbreak-specific effect on four things an analyst can see before fitting — dispersion, total cases, series length and mean weekly count — explains 34% of the variation for the observation model and a median of **1.8%** for the other five. Of 24 correlations, 5 reach $p < 0.05$ and 3 survive control of the false-discovery rate, all of them for the two factors that have a mechanism.

Table 4: How much of each factor’s outbreak-to-outbreak variation is predictable from the outbreak, by ordinary least squares on four pre-fit descriptors.

Factor	Variation explained	SD of the effect
Observation model	34.3%	0.291
Fixed parameters	3.5%	0.029
Temperature form	2.0%	0.152
Model structure	1.6%	0.111
Fraction of series fitted	1.0%	0.238
Rainfall lag	0.1%	0.185

The rainfall lag is the clearest case: it moves the verdict in one outbreak in six, its effect varies as much across outbreaks as the temperature form’s, and *none* of that variation is predictable from anything measurable beforehand. An analyst cannot know whether their lag choice matters without trying the others.

What survives, and is the stronger statement, is that the field’s habit of reaching first for a contextual explanation reaches for a term bounded at a quarter, while three-fifths of the variation is not a property of the setting or of the method but of what happens when a particular method meets a particular epidemic. That term is invisible to both of the standard remedies. It is the reason the recommendation this paper ends on is an interval computed within each outbreak rather than a house style.

The published moderator does not reproduce here. The specific contextual explanation on offer is that the association is strongest where temperature varies most [5]. We tested its closest analogues on our own data — six descriptions of each outbreak’s temperature regime against three outcomes: the estimated Brière exponent, the share of analyses endorsing climate forcing, and the disagreement between two of them. Eighteen correlations, so about one would reach $p < 0.05$ by chance and reporting raw significance would itself be the practice this paper measures.

Seven reach $p < 0.05$, against about one expected by chance, so the temperature regime is not unrelated to what the analyses find. But every association is weak — no $|\rho|$ exceeds 0.22 — and **one survives control of the false-discovery rate**: the estimated temperature

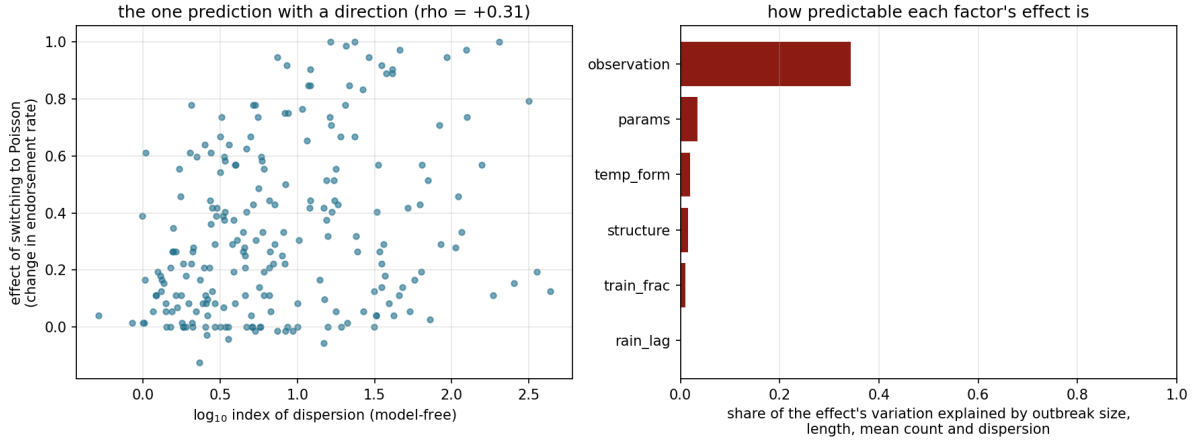


Figure 2: Left: the one prediction made in advance — the effect of assuming Poisson counts against the model-free index of dispersion. Right: how much of each factor’s outbreak-to-outbreak variation four pre-fit descriptors account for. Only the factor with a mechanism behind it is predictable, and only in part.

exponent against the season’s temperature range, $\rho = -0.22$ ($q = 0.02$), whose sign is the wrong way round for the hypothesis, the exponent being *smaller* where temperature varies more. The endorsement rate moves in the expected direction but does not survive correction ($\rho = +0.15$, $q = 0.10$). Both facts belong in the same sentence: there is something there, and it is small.

We do not read this as refuting the published result, which was computed on a different quantity: they measured the strength of a statistical correlation between incidence and temperature, we the exponent of a mechanistic response. What it does establish is that the moderator is not recoverable from a design in which the analysis is held fixed — which is what it would have to be if it were a property of dengue rather than of the estimates that reached print.

A mechanistic refinement we expected to do better also failed. Transmission responds to temperature along a unimodal curve, so a season spent near the flat optimum should offer nothing to detect while one crossing the steep flank offers a great deal; the mean gradient traversed should therefore predict what the data can support. It does not ($\rho = +0.13$, $q = 0.11$), and it is reported because a study about undisclosed analytical choices does not get to drop the measure it hoped would work.

4.2 What this does to the number the field reports

The verdict on climate forcing is a yes or a no, and a reader may reasonably feel that a yes/no is a coarse thing to be unstable about. The same decomposition can be applied to the quantity these papers actually report: the basic reproduction number.

Table 5: Variation in the estimated R_0 of the same 221 outbreaks. Reported under four treatments of the long right tail, because the contrast and not the exact figure is the finding.

Treatment	Constant model		Climate model	
	Between	Within	Between	Within
Raw	72.8%	27.2%	2.1%	97.9%
Log scale	71.6%	28.4%	19.8%	80.2%
Restricted to $R_0 \in [0.5, 20]$	72.8%	27.2%	17.1%	86.3%
Restricted to $R_0 \leq 10$	72.8%	27.2%	21.1%	81.4%

The constant model’s R_0 is mostly about the epidemic; the climate model’s is mostly about the analyst. Under the constant model roughly 72% of the variation in estimated R_0 lies between outbreaks — the estimate is measuring something that differs between epidemics, which is what it is for. Under the climate model that falls to between 2% and 21% depending on how the long right tail is handled, and the analytical share rises to 80–98%. The contrast survives every treatment we tried and is far larger than the differences between them.

Put on the scale a reader can weigh: the median outbreak’s R_0 estimate ranges over **1.31** across the 144 analyses, against a median estimate of **1.29**. The spread induced by the choices is about the size of the quantity being estimated.

The three-way partition of Section 4.1 says something the two-way one cannot, and it is the practical question: could agreeing on a method fix this?

Table 6: Variation in estimated R_0 , partitioned into what a convention could reach (the analysis) and what it could not (the interaction). On the log scale, which handles the long right tail; the raw scale is more extreme in the same direction (0.5% analysis, 97.4% interaction under the climate model). The climate-model row uses the 219 outbreaks whose R_0 is positive and finite under every analysis, which is why its “outbreak” share reads 22.5% here against 19.8% in the table above, computed on all 221.

	Outbreak	Analysis	Interaction
R_0 , constant model	71.6%	5.3%	23.0%
R_0 , climate model	22.5%	8.4%	69.1%

No convention would fix it. The analysis main effect for R_0 is about a twentieth under either model — there is no systematically high-reading method to stop using. What adding climate terms does is move variation out of the outbreak and into the *interaction*, which rises from 23% to 69%. The instability in the number this field reports is not a house style anyone could adopt their way out of; it is specific to the meeting of a method with an epidemic, and the only report that carries it is one made within the outbreak.

This connects to the identifiability result in the appendix from the other direction. There, adding climate parameters to a short series destroyed the identifiability of the population at risk, which the simpler model recovered to within a factor of two. Here the same addition destroys the stability of R_0 . The climate terms are not free: they are paid for in the precision of everything else in the model, and that cost is not usually reported because it is not usually looked for.

4.3 How often the verdict changes, measured two ways

Table 7: Stability of the in-sample verdict within each outbreak.

	Windows	Share
Always favours climate forcing	18	8.1%
Never favours climate forcing	0	0.0%
Verdict changes with the choices	203	91.9%

Bootstrapping over windows, the independent units, gives a 95% interval of 87.8–95.5%. Among unstable windows the median split is 107 of 144 combinations in favour. The instability appears in every well-represented country, ranging from 85% to 100%.

That figure is not comparable across studies, and we do not rely on it. Asking whether *any* two combinations disagree becomes easier the more combinations there are: a

design with 144 cells will exceed one with 24 even if nothing about the evidence has changed. Since this study grew its own design twice, the point applies to it directly.

We therefore report alongside it the probability that two analyses of the same outbreak, drawn at random, reach opposite verdicts. With p the proportion of analyses favouring climate forcing, that is $2p(1 - p)$: a function of the proportion and not of how finely the space was enumerated, zero for a unanimous outbreak and one half for an evenly split one. Across the 221 outbreaks it is **29.8%** (median 34.6%), and it exceeds 25% in 61.1% of them.

Table 8: Every design this study ran. The share of outbreaks where some pair disagrees climbs with the number of cells; the pairwise probability does not.

Design	Combinations	Outbreaks	Some pair disagrees	P(two disagree)
Four factors	24	236	82.6%	29.9%
Five factors	48	234	88.0%	31.4%
Six factors	144	221	91.9%	29.8%

Each factor was added to close a gap the previous version had named in its own limitations section, not sought once a result was known; the fixed epidemiological parameters, added for that reason, turn out to be the weakest factor of the six. The invariant statistic is flat across all three designs at about 30%, which is the figure we ask readers to carry away.

4.4 Statistical conventions dominate; the biology does not

Table 9: Effect of each choice, paired within window so that differences in series length or case count cannot drive the comparison.

Choice varied	Verdicts flipped	P(climate wins) before → after
<i>Statistical conventions</i>		
Negative binomial → Poisson	34.3%	0.570 → 0.900
75% → 100% of the series	20.7%	0.702 → 0.769
<i>Covariate handling</i>		
Rainfall lag 3 → 7 weeks	15.8%	0.734 → 0.738
Brière → log-linear temperature	11.5%	0.696 → 0.774
Rainfall lag 3 → 5 weeks	11.0%	0.734 → 0.733
<i>Epidemiological content</i>		
Host–vector → human-only SEIR	9.9%	0.738 → 0.727
Host–vector → thermal mortality	9.4%	0.738 → 0.741
Central → alternative fixed parameters	3.1%	0.735 → 0.736

The factors fall into three groups, and the ordering is the result. The *statistical conventions* — which likelihood, how much of the series to fit — flip 21% and 34% of verdicts. The *covariate handling* follows at 11–16%. The *epidemiological content* — which transmission mechanism, and what the biological parameters are — comes last at 3–10%. On this evidence the question is answered by the statistics rather than by the biology.

The two factors added specifically to answer the objection that the study was testing a toy model are among the weakest. Replacing the host–vector formulation with a directly transmitted SEIR flips 9.9%; replacing it with a host–vector model whose mosquito mortality follows temperature flips 9.4%. Moving the fixed epidemiological parameters — lifespan, incubation periods, vector-to-host ratio — to the other end of their published ranges flips **3.1%** and moves the endorsement probability by 0.001. We expected that factor to matter and it does not.

Assuming Poisson counts raises the probability of endorsing climate forcing from 0.57 to 0.90. Two mechanisms could produce that, and they make opposite predictions, so we measured rather than argued.

The first is the one usually given, and the one an earlier draft of this paper gave: Poisson treats ordinary noise as a large residual, the fit chases individual weeks, and the climate covariates supply the flexibility to do it. If so their coefficients should be pushed harder under Poisson. **They are not.** Paired within outbreak and within every other choice, the median $|a_{\text{temp}}|$ is 0.092 under a negative binomial and 0.097 under Poisson, and is larger under Poisson in 40% of 15,912 pairs — less often than chance. For $|a_{\text{rain}}|$ it is 46%. The climate model does the same thing under both likelihoods.

What changes is what the criterion pays for it. The climate model nests the constant one, so it fits at least slightly better always; the question is only whether the improvement clears two parameters' worth of penalty. Without a dispersion parameter to absorb the residual, the same improvement buys far more log-likelihood: the median ΔAIC is -1.6 under a negative binomial and -68.9 under Poisson, a ratio of **13** where the two agree in sign, and larger in magnitude under Poisson in 90% of pairs. The asymmetry is near-total: climate forcing wins under both in 56.4% of pairs, under Poisson alone in **33.6%**, and under the negative binomial alone in **0.6%**.

So the mechanism is not that Poisson makes the model chase noise. It is that Poisson makes the criterion generous, and it is generous in one direction only. The dispersion it ignores is real, and is expected: infectious disease counts are overdispersed by individual variation in transmission [18], and the standard remedies for it in count regression are long established [19]. Taking residuals about a five-week moving average — a measure independent of any transmission model — these series have a median index of dispersion of **5.7**, quartiles 2.6 and 19.1, and a quarter above 20. A week of 2,000 cases that Poisson treats as accurate to ± 45 is accurate to about ± 107 .

A reader may object that Poisson is not a defensible choice at all on overdispersed counts, and that including it inflates the instability by admitting a cell that should never have been in the multiverse. Two things answer this. The field uses it: the systematic review of 99 dengue prediction models finds 18.3% using Poisson regression and 18.3% linear regression, with the negative binomial not appearing as a category of its own [1], and a head-to-head comparison on mosquito-borne outbreak data finds the negative binomial the better of the two [9]. That is a large minority rather than a majority, which is the honest reading and still enough that a multiverse excluding Poisson would not describe the literature it is about. And the result does not depend on it. Restricting the entire study to negative-binomial fits — discarding half the factorial as indefensible — still leaves **79.6%** of outbreaks changing verdict (95% CI 75–85), because the remaining five choices continue to move it. The observation model is the largest lever, not the only one.

The rainfall lag behaves differently and is worth separating. It flips one verdict in six while moving the average by 0.004. That is variance injected into a conclusion by an arbitrary choice, without any systematic effect to justify it.

4.5 The specification curve

Figure 4 shows the multiverse directly, in the two forms the question takes.

Within one outbreak. The left panel draws the most evenly split window in the study — Jamaica, July 2023 — where exactly 72 of the 144 specifications favour climate forcing and 72 favour the constant model. The ΔAIC values run from -337 to $+4$. One analyst, following defensible conventions, would report overwhelming evidence for climate-driven transmission in this epidemic; another, following different but equally defensible conventions, would report no evidence at all. The dot matrix below shows why: the observation-model row splits the panel exactly, every Poisson specification lying on the climate side and every negative binomial on the other.

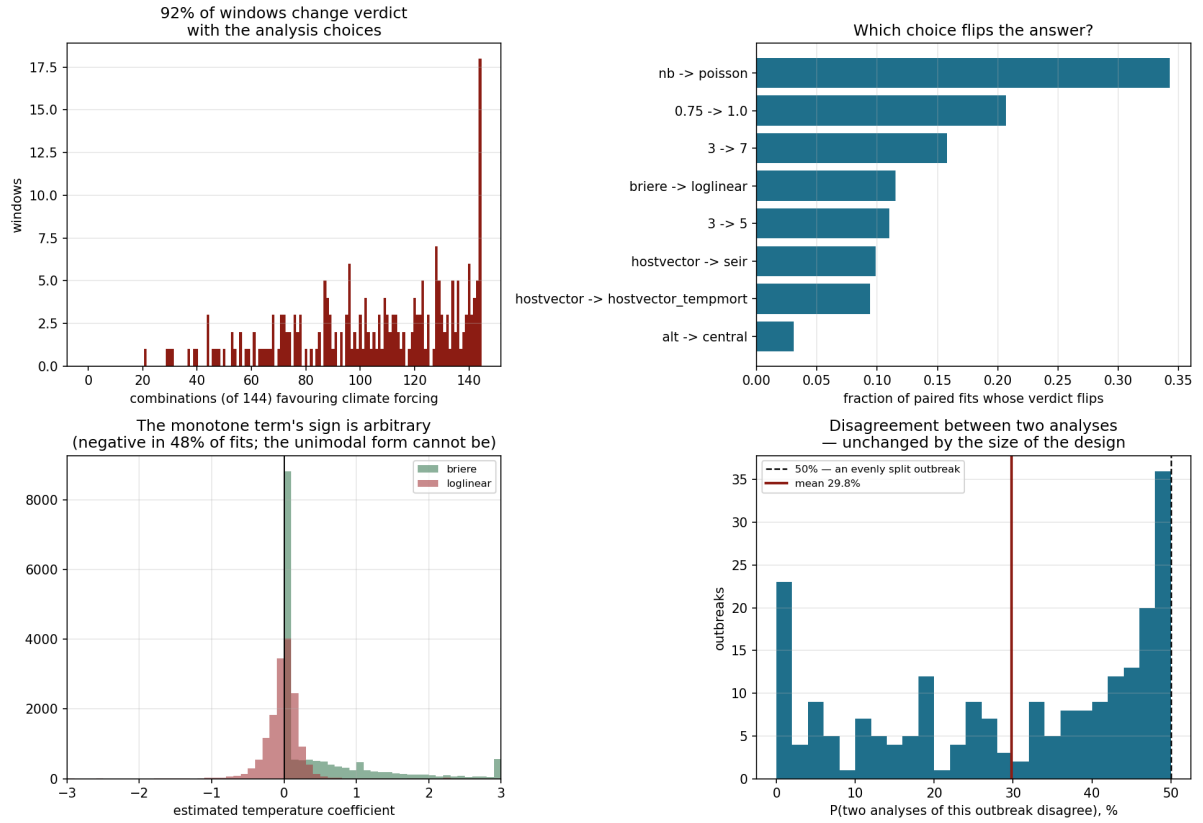


Figure 3: The four results of the factorial. Top left: how many of the 144 analyses favour climate forcing, per outbreak — the mass sits well short of unanimity at either end. Top right: the share of paired comparisons each choice flips, which separates the statistical conventions from the epidemiological content. Bottom left: the estimated temperature coefficient by parameterisation, clipped to ± 3 ; the monotone form is negative in half of all fits and the unimodal form cannot be. Bottom right: the probability that two analyses of one outbreak disagree, the statistic that does not grow with the design.

Across outbreaks. The right panel gives each specification the whole study and asks how often it endorses climate forcing. If the choices were interchangeable the result would be a flat line. The most credulous specification — Poisson counts, a log-linear temperature term, a three-week rainfall lag, the first 75% of the series, an SEIR structure and the central parameter set — endorses climate forcing in **95.9%** of outbreaks. The least credulous differs from it in exactly two of the six, the observation model and the temperature form: negative binomial and Brière, with the same lag, the same fraction, the same structure and the same parameters. It endorses climate forcing in **39.8%**. That is a spread of **56 percentage points** between two analyses of the same data, neither of which any reviewer would reject on its face.

This is the finding stated as a rate rather than as a count of flips, and it is the form in which it bears on reading the published literature: a paper reporting that climate forcing improved the fit has told the reader where in this range its analysis sat only if it stated all six choices, which such papers generally do not.

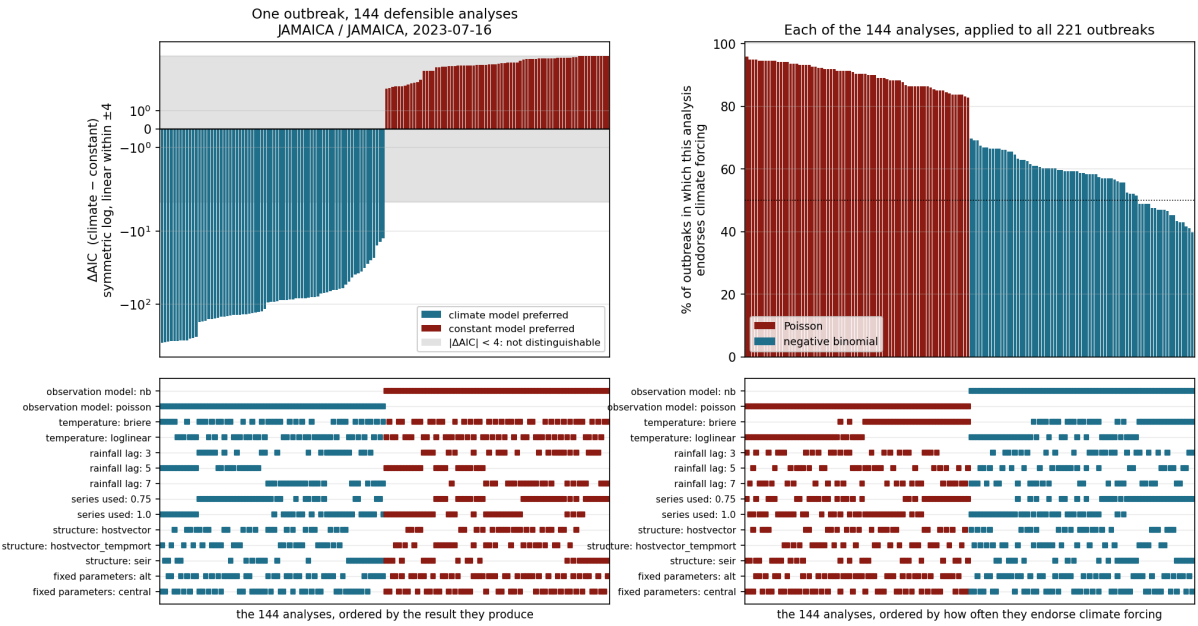


Figure 4: The specification curve. Left: all 144 analyses of a single outbreak, ordered by the result they produce, on a symmetric-log scale that is linear within ± 4 so that both directions remain visible. Right: each of the 144 analyses applied to all 221 outbreaks, ordered by how often it endorses climate forcing. Lower panels mark which choices each analysis used.

4.6 Out-of-sample validation halves endorsement but not instability

On three Pakistani windows examined first, held-out deviance preferred the constant model in every configuration, suggesting that validation was the stable criterion. Across 221 outbreaks that does not hold.

Table 10: In-sample and out-of-sample compared on the same fits.

	Unstable within window	Climate preferred
In-sample (AIC)	84.7%	70.3% of fits
Out-of-sample (held-out deviance)	90.6%	36.3% of fits

Validation halves the rate at which climate forcing is endorsed — a large and useful reduction in false positives — but leaves the verdict at least as dependent on the remaining choices:

90.6% of outbreaks are unstable out-of-sample against 84.7% in-sample. It is a corrective, not a solution. Earlier versions of this study, on smaller designs, found the two rates equal; enlarging the design moves the out-of-sample figure above the in-sample one, which is the opposite of what a remedy would do.

4.7 The conventional temperature term reports an arbitrary sign

Table 11: Estimated temperature coefficient by parameterisation, 15,912 fits each.

Parameterisation	Mean	Median	Negative in
Brière (unimodal, from vector thermal biology)	+0.583	+0.021	0.0%
Log-linear (conventional)	−0.000	+0.000	48.5%

The log-linear coefficient is negative in essentially half of all fits and positive in the rest, averaging to nothing. The explanation usually offered is that a monotone term fitted to a season whose temperature falls while cases rise must return a negative coefficient whatever the biology, so the sign is a property of the calendar. We tested it rather than repeating it. Against the temperature trend over each outbreak’s growth phase the estimate correlates at $\rho = +0.25$ ($p = 1 \times 10^{-4}$): where temperature falls during growth the coefficient is negative in 52% of outbreaks, where it rises, in 32%.

So the direction is as claimed and the size is not. The calendar tilts the sign without setting it, which leaves most of the sign unexplained — and a coefficient whose sign is half calendar and half unaccounted for is not a measurement of anything. A unimodal response cannot produce a negative value at all, because it has no direction to choose.

4.8 A remedy, and it is nearly free

Table 12: Candidate decision rules. Both columns matter: a rule that is never wrong because it never speaks is not a remedy.

Rule	Some pair disagrees	95% CI	P(two disagree)	Answers for
Sign of ΔAIC (conventional)	91.9%	88–96%	29.8%	100%
$ \Delta\text{AIC} > 2$	89.6%	86–93%	24.9%	100%
$\Delta\text{AIC} > 4$	39.8%	33–45%	4.0%	100%
$ \Delta\text{AIC} > 10$	37.0%	32–44%	4.5%	97.7%
$ \Delta\text{AIC} > 20$	34.9%	29–41%	4.7%	95.9%
Negative binomial only	79.6%	75–85%	—	100.0%
Negative binomial, $ \Delta\text{AIC} > 10$	20.0%	14–25%	—	74.7%
All 144 combinations must agree	0.0%	—	—	8.1%

Read the two instability columns together, because separately each misleads. Between margins of 2 and 4 the *pairwise* disagreement collapses from 24.9% to **4.0%** while the rule still answers for every outbreak: on that scale the remedy works exactly as it did on the smaller designs, and at the same place. The share of outbreaks in which *some* pair still disagrees falls only from 90% to 40%, but that column is the one that grows with the size of the design, and it is not comparable with the 6.9% obtained from 48 combinations.

Both are true and the pair is the finding. Where dissent survives a margin of 4, the dissenting minority is small: a median of 3 of the 100 analyses still speaking, 2.8% of them. A margin makes the surviving disagreement rare within an outbreak without making it absent from the study.

The interpretation is direct: **almost all of the instability lies in comparisons where $|\Delta\text{AIC}| < 4$** , that is, where the models were never distinguishable. Burnham and Anderson’s long-standing guidance treats ΔAIC below 4 as substantial support for both models [3]. The recommendation is therefore not new; what is new is the measured cost of not following it.

Two alternatives fare worse. Requiring unanimity across the factorial is perfectly stable and speaks for one outbreak in twelve. Fixing the observation model alone — the single largest lever — still leaves 80% of outbreaks with some disagreement, because the remaining choices continue to move the verdict. Combining the two does better: restricting to the negative binomial and requiring $|\Delta\text{AIC}| > 10$ leaves 20%, and speaks for three outbreaks in four.

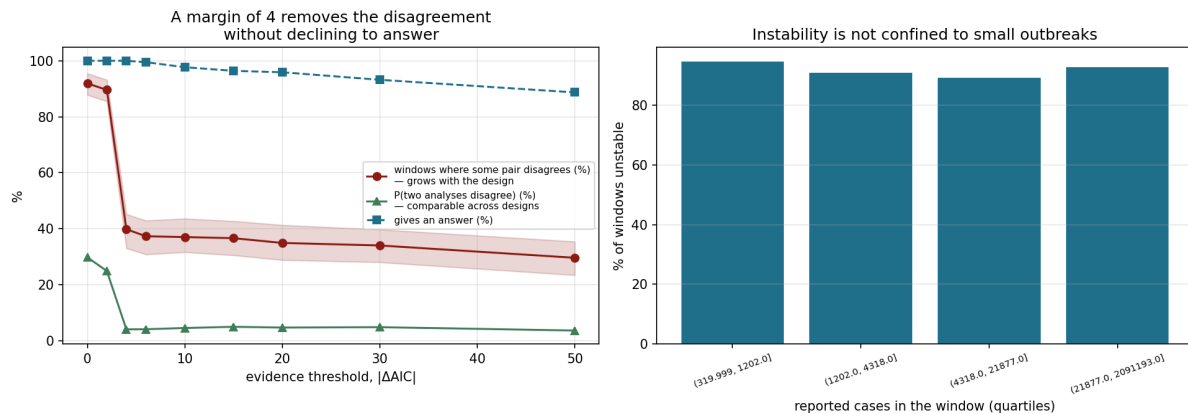


Figure 5: Left: the effect of an evidence margin on both scales, with the share of outbreaks still receiving a verdict. The share where some pair disagrees (dark) and the probability that two analyses disagree (green) both collapse between 2 and 4, but to very different levels, and only the second is comparable with the smaller designs. Right: instability by quartile of reported cases.

4.9 The instability is not specific to AIC, and stricter criteria are worse

AIC, BIC and the likelihood-ratio test are all thresholds on the same quantity, placed at different points, so all three follow from what was already computed: $\text{AIC} = -2 \log L + 2k$ recovers the log-likelihood, and $\Delta k = 2$ in every comparison.

Table 13: Instability under each criterion, used conventionally. All are thresholds on ΔAIC ; the second column states where each one falls.

Criterion	Equivalent to $\Delta\text{AIC} <$	Unstable	95% CI
AIC, sign	0.00	91.9%	88–96%
Likelihood-ratio test, $p < 0.05$	−1.99	95.0%	91–98%
BIC, sign (at the median 38 weeks)	−3.28	96.8%	95–99%
Likelihood-ratio test, $p < 0.01$	−5.21	97.3%	95–99%
AIC, abstain within ± 4	—	39.8%	33–45%
BIC, abstain within ± 4	—	90.5%	86–94%

Strictness makes matters worse, monotonically. The significance test an epidemiologist would actually run is equivalent to a margin of about 2 and leaves 95% of outbreaks unstable. A margin of 4 on the BIC scale, which sounds like the same remedy, leaves 91%, because BIC’s additional penalty shifts the abstention band off centre — it abstains over $(-7.3, +0.7)$ rather than $(-4, +4)$, so one side of the disagreement is still allowed to speak. The distinction that matters is not how strict a rule is but whether it abstains symmetrically about zero.

4.10 Why: large movement, marginal dissent

Two facts hold simultaneously and sound contradictory until separated.

The choices move the evidence enormously. The range of ΔAIC across the 144 combinations of a single outbreak has a median of **804 AIC units** (quartiles 140 to 11,105). The median $|\Delta\text{AIC}|$ being interpreted as evidence is 14.4. The analyst’s own choices move the evidence by roughly **fifty-six times** the size of the difference being read from it.

Yet the dissent is marginal. The margin needed to eliminate all disagreement within a window has a median of 4.00, and in **60.2%** of outbreaks every dissenting combination does so on evidence weaker than $\Delta\text{AIC} = 4$.

Here the six-factor design corrects a claim the smaller ones supported. On 48 combinations the margin needed was uncorrelated with how far the choices moved the evidence, and dissent looked like a pure property of comparisons that were weak to begin with. On 144 the correlation is $\rho = +0.61$ ($p = 4 \times 10^{-24}$): in the remaining 39.8% of outbreaks at least one dissenter is confident, and no band around zero can reach it. What stays true is that such dissenters are few — a median of 3 of the 100 analyses still speaking.

Both hold because most of that enormous variation stays on one side of zero: it changes how strong the evidence appears without changing which model it favours. In three outbreaks in five, everything that does cross zero crosses it weakly, and a band around zero reaches all of it. In the other two, something crosses decisively and no band will ever reach it — but when that happens the dissenter is nearly alone, a median of 3 of the 100 analyses still speaking.

That is why a margin of 4 removes almost all pairwise disagreement (29.8% to 4.0%) while leaving 39.8% of outbreaks with at least one dissenting cell. The two figures describe the same fact from opposite ends and neither alone is honest: the surviving disagreement is common across outbreaks and rare within them.

It also explains why a stricter threshold cannot help. A stricter rule is still a rule with a boundary, and the confident dissent sits far from any boundary one might choose; moving the boundary only changes which of the weak comparisons are silenced.

We also tested and rejected a simpler explanation — that instability tracks the density of comparisons at the boundary. Scanning the threshold from -14 to $+2$ gives instability that runs backwards rather than forwards against the local density ($\rho = -0.27$), so the hypothesis is not merely unsupported: it predicts the wrong sign.

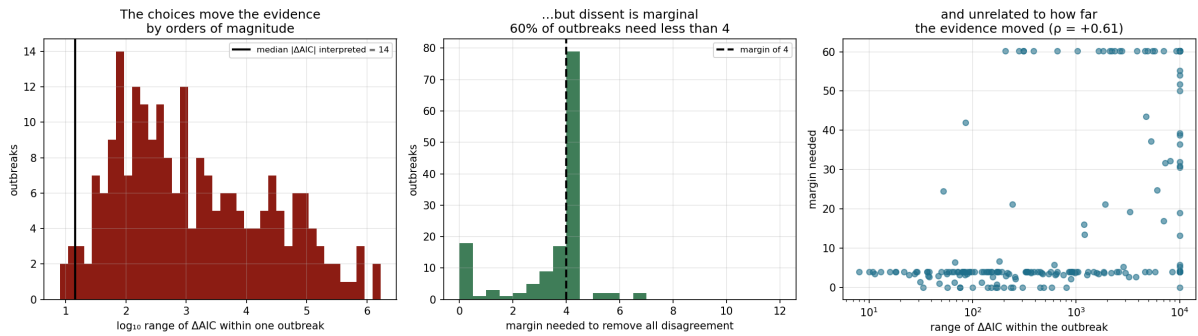


Figure 6: Left: the range of ΔAIC induced within a single outbreak by the analysis choices, against the median difference interpreted as evidence. Centre: the margin needed to remove all disagreement within a window. Right: the two are unrelated.

4.11 How often is the verdict wrong, and in which direction?

Everything above measures disagreement, which shows that something is wrong without showing which answer is wrong: on real data there is no truth to compare against. We therefore simulated

counts from models whose answer is known, using each window’s own fitted parameters, its own climate series and its own estimated dispersion, and ran the full factorial on them. Eighty windows, two truths, all 144 combinations: 23,000 fits.

The two truths are matched on mean transmission. Switching the climate coefficients on multiplies $\beta(t)$ by a factor whose average is not one, so an unadjusted alternative would differ from the null in overall transmission intensity as well as in whether transmission varies — and in an earlier version of this analysis the raised R_0 made the generating integration diverge for 21 of 59 windows, so power was estimated only where a strong effect happened not to break the model. Rescaling β_0 by the mean of the climate multiplier isolates the contrast the study is about and leaves both arms covering the same windows.

Table 14: False positives (data generated with no climate effect) and power (data generated with an effect of realistic size). Separation is power minus false-positive rate: how far the rule distinguishes the two truths, as opposed to how often it says yes.

Observation model	Rule	False positives	Power	Separation
Poisson	sign of ΔAIC	85.7%	95.3%	10 pts
Poisson	margin 4	78.6%	93.5%	15 pts
Poisson	margin 24 (its best)	58.8%	86.6%	28 pts
Negative binomial	sign of ΔAIC	18.7%	73.4%	55 pts
Negative binomial	margin 4	6.2%	61.8%	56 pts
Negative binomial	margin 12	3.3%	43.3%	40 pts

A Poisson observation model does not discriminate. On data containing no climate signal it endorses climate forcing six times in seven. Its 95% power is not a virtue next to an 86% false-positive rate: the procedure is agreeing with almost everything. Over every margin tested its best separation is 28 points, and reaching even that requires abstaining below $|\Delta\text{AIC}| = 24$.

The observation model is worth 45 points of separation. Switching to a negative binomial moves separation from 10 to 55 at the identical decision rule. Nothing else examined here approaches that.

The margin is a specificity dial, not an improvement. On the negative binomial it trades power against false positives along the same curve rather than shifting the curve outward — separation is 54.7 at margin 0 and 55.6 at margin 4, essentially unchanged while false positives fall from 18.7% to 6.2%. It is the right control when a false positive costs more than a miss, and it is not a substitute for the observation model: applied to Poisson counts a margin of 4 still leaves 79% false positives.

This bears on the empirical result, but less than an earlier draft claimed. Poisson endorsed climate forcing in 90.0% of real fits and the negative binomial in 57.0%. Pairing the two within outbreak and within every other choice, Poisson endorses where the negative binomial does not in **33.6%** of real pairs; that same discordant cell occurs in **67.1%** of simulated pairs when no climate effect exists and **23.3%** when one does. The cell is diagnostic — three times as common under the null — but the real rate sits nearer the arm that contains an effect than the arm that does not. The earlier draft stated that 85.7% of Poisson endorsements would occur on data with no effect, and that most of the gap between the observation models was therefore not detection. Neither follows: 85.7% is the false-positive rate under a null truth and not a share of endorsements, and a mixture read off these three rates puts the null fraction nearer a quarter than a majority. We report the rates and decline the inference, for the same reason given below — the simulated effect size comes from these same fits.

The instability is not manufactured by running 144 analyses. The arithmetical objection to the headline is that with enough analyses something will always disagree. The simulation

answers it, because here the truth is known and the factorial is identical in both arms. Measured on the scale that does not grow with the design, two analyses of the same simulated outbreak reach opposite verdicts **44.5%** of the time when no climate effect is present — close to the 50% ceiling, which is what pure coin-flipping looks like — and **18.6%** when a real effect of realistic size is. The truth more than halves the disagreement, so the design is not generating it out of nothing.

The more uncomfortable half is the 18.6%. Even when a genuine climate effect is present, at a size drawn from these epidemics’ own fitted parameters, two defensible analyses still reach opposite conclusions almost one time in five. The instability is not only a symptom of weak evidence; a real signal of ordinary size does not survive it either.

The remedy holds under both truths. Applying the recommended rule to the same simulated data — abstain within $|\Delta\text{AIC}| = 4$ — reduces pairwise disagreement from 44.5% to **1.8%** where no climate effect exists, and from 18.6% to **4.3%** where one does. It works in both cases and works slightly less completely where an effect exists, which is the expected direction: a real effect produces large $|\Delta\text{AIC}|$, so some dissent falls outside the abstention band. A margin removes disagreement about comparisons too close to call; it cannot remove disagreement about a difference the analyses genuinely disagree about.

Which arm do the real outbreaks resemble? A one-number answer cannot say. It is tempting to compare the real data with the two simulated arms, and two attempts to do so failed. On pairwise disagreement the real outbreaks sit between (29.8%, against 44.5% with no effect and 18.6% with one); under the margin-4 rule the ordering reverses (4.0%, against 1.8% and 4.3%). An earlier version of this study, using the design-dependent share of unstable windows, reported that the real data resembled the no-effect arm; that reading did not survive the invariant measure and was withdrawn.

The reason both attempts failed is now visible: the arms differ in three components and each of those statistics compresses them to one. Partitioning the same three datasets the same way separates them.

Table 15: The three-way partition applied to both simulated arms and to the real outbreaks. Same windows, same 144 analyses, same fitted parameters; only the truth differs between the first two rows.

	Outbreak	Analysis	Interaction
Simulated, no climate effect	10.9%	47.0%	42.1%
Simulated, real effect present	28.6%	10.3%	61.1%
Real outbreaks	23.5%	15.9%	60.6%

The analysis main effect is what discriminates, and it does so sharply. Where there is nothing to detect it is the largest of the three at 47%, which is what should happen: with no signal the answer is whatever the convention gives, and the convention does not vary by window. Where a real effect is present it collapses to 10%. The real outbreaks read **15.9%** — with the effect arm, four times away from the null arm — and their outbreak and interaction shares land within five points of that arm on both.

We report this as evidence and not as proof. The simulated effect size is drawn from these same fits, the two arms are not a calibrated mixture of what the world contains, and a fingerprint matching one arm does not establish that every outbreak in it carries a signal. What it does establish is that the real data does not carry the signature of a literature detecting nothing, which is the reading the earlier withdrawn sentence had pointed toward. It is also consistent

with Section 4.13, where our own rule gives a clear verdict for 137 outbreaks and 86% of those support climate forcing.

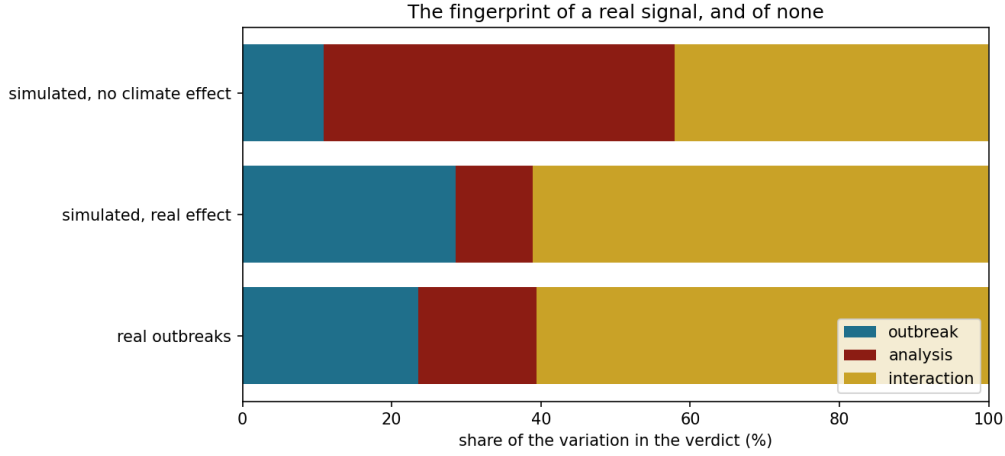


Figure 7: The fingerprint of a real signal and of none. The analysis main effect (centre band) is the discriminating component: largest where there is nothing to detect, small where there is something.

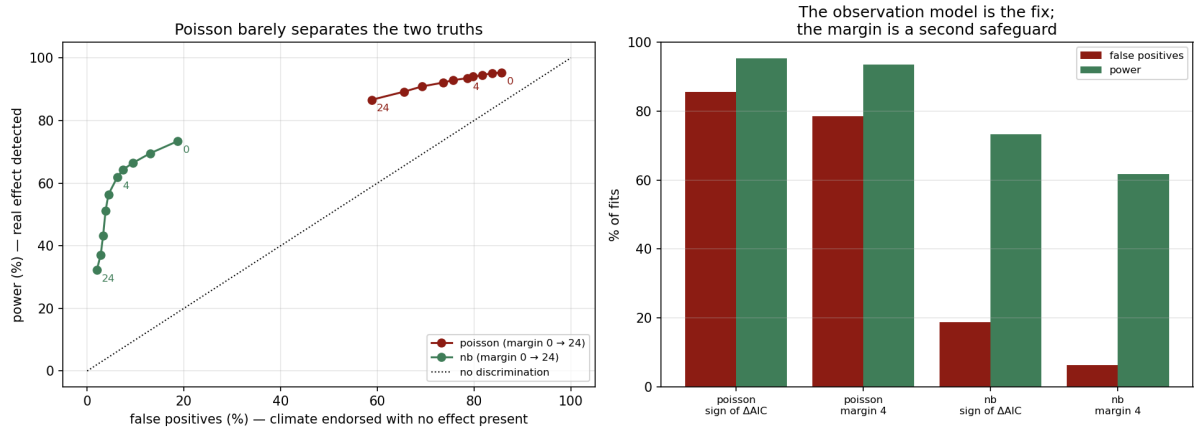


Figure 8: Left: power against false positives as the decision margin varies from 0 to 24. The Poisson curve lies close to the diagonal of no discrimination. Right: the same four rules as rates.

4.12 The intervals do not cover, and an interval that does

Everything above concerns which model is chosen. A paper also reports a number with an interval around it, and that interval makes a specific promise: that it contains the true value 95 times in 100. The promise is testable here, because the truth is set by us.

Counts were simulated from each window's own fitted parameters with the Brière temperature exponent set to 1.0 and the rainfall coefficient to 0.30, and every Brière combination was asked to recover them — 80 outbreaks, 72 combinations each, 5,760 fits. A conventional analysis reports one estimate and $\hat{\theta} \pm 1.96 \text{ SE}$ from one set of choices. Coverage is computed over every cell, so the figure describes what an analyst drawing any defensible set of choices would obtain, not one arbitrary choice.

A conventional 95% interval for the rainfall coefficient contains the truth 71% of the time. For the temperature exponent it manages 83%, and for the transmission coefficient

Table 16: Coverage of a nominal 95% interval. The multiverse interval combines all analyses of an outbreak by Rubin’s rules, whose within- and between-analysis variance decomposition applies unchanged here: several analyses of one dataset, each with its own estimate and its own sampling variance.

Interval	Temperature exponent	Rainfall coefficient	Transmission coefficient
Conventional, one analysis	82.9%	71.2%	78.0%
Multiverse (Rubin)	100.0%	97.5%	100.0%

— which fixes R_0 within a structure, so the two have the same coverage — **78%**. All three are below the promise, and the shortfall is not a subtlety of the asymptotics: a conventional interval reports the square root of the within-analysis variance and behaves as though the between-analysis variance were zero, when for the rainfall coefficient the two are of the same size.

The point estimates are no better behaved. The ratio of the estimated transmission coefficient to the truth has a median of 1.27 and an interquartile range of 0.74 to 2.59: in the middle half of analyses the reported R_0 is between three-quarters and two and a half times the value that generated the data. That is the number this field reports, on data whose answer we set ourselves.

Combining the analyses restores the promise. Rubin’s rules [20] give $\bar{\theta} \pm t\sqrt{W + (1 + 1/m)B}$, with W the mean within-analysis variance and B the variance between the estimates. Coverage becomes 97.5% and 100.0%, at the cost of an interval 1.5 to 2.2 times wider. The 100% is conservative rather than exact and we report it as such; with $m = 72$ analyses and a long-tailed estimate distribution, B is generous.

That trade is the whole recommendation in one line. The narrow interval is not more informative than the wide one; it is the same information with a promise attached that it does not keep.

Eight analyses are enough. A recommendation costing 288 fits per outbreak would be admired and not used, so we asked how far it can be cheapened. Drawing m analyses at random from an outbreak’s 72 and combining only those:

Table 17: Coverage of the multiverse interval against the number of analyses combined, averaged over 20 random draws per outbreak.

Analyses combined	2	4	8	16	24	72
Rainfall coefficient	91.0%	96.2%	97.3%	97.6%	97.6%	97.5%
Temperature exponent	94.9%	98.6%	100.0%	100.0%	100.0%	100.0%
Transmission coefficient	91.7%	97.1%	98.9%	99.9%	99.9%	100.0%

Four analyses already reach nominal coverage and eight are indistinguishable from seventy-two. The interval also stops narrowing after eight (median half-width 0.545 against 0.542 at 72), so nothing is bought by going further. Eight fits of a climate model and eight of a null is a few minutes of computation on one processor, which is a different proposition from a factorial and is what we would ask of a practitioner.

Which eight, and a result we had assumed the other way round. A practitioner has to choose the eight, so we compared a spanning draw against draws holding one factor fixed.

We had expected the observation model to be the choice that must be varied, since it is by far the largest lever on the verdict. It is close to irrelevant to the interval: eight analyses that all use the same likelihood cover as well as eight that do not. What cannot be held fixed is the *rainfall lag*, and only for the *rainfall* coefficient, which it loses nine points.

Table 18: Coverage of a nominal 95% interval from eight analyses that share one choice. Only the factor defining a covariate matters for that covariate’s coefficient.

Eight analyses sharing...	Rainfall coef.	Temperature exp.	Transmission coef.
nothing (a spanning draw)	97.5%	99.9%	99.2%
one observation model	97.1%	98.9%	98.6%
one model structure	97.9%	99.7%	96.5%
one fixed parameter set	97.4%	99.7%	98.6%
one fraction of the series	97.0%	99.6%	99.2%
one rainfall lag	88.7%	97.7%	97.8%

The rule that fits is narrower and more useful than the one we assumed: **vary the choices that construct the covariate whose coefficient you are reporting**. The lag is what makes the rainfall covariate; hold it fixed and the analytical uncertainty in its coefficient is never entered into the arithmetic, however many other things are varied. A practitioner reporting a rainfall effect needs three lags and can fix everything else; one reporting a verdict needs both observation models, which is a different requirement for a different quantity.

And when the data did not come from any fitted model. The objection to everything above is that it is circular: the counts were simulated from the family that is then fitted, and real surveillance is not generated by a host–vector SEIR with a Brière response. We therefore repeated it with a generator outside the family, and the mis-specification is not arbitrary — the appendix demonstrates it in these data. Provincial epidemics are sums of district epidemics that peak at different times, and a homogeneously mixing model reads the broader sum as slower transmission. So counts were generated from *two* climate-forced patches sharing one temperature series and one exponent, seeded 46 days apart (the offset the two-patch model infers for Khyber Pakhtunkhwa), and every fitted model has one patch.

Table 19: Coverage of a nominal 95% interval when the generating model is outside the fitted family. 40 outbreaks, 2,700 fits.

	Temperature exponent	Rainfall coefficient
<i>Data from the fitted family</i>		
Conventional	82.9%	71.2%
Multiverse	100.0%	97.5%
<i>Data from two patches, fitted with one</i>		
Conventional	78.9%	42.0%
Multiverse	95.0%	90.0%

The conventional interval degrades to 42%. A 95% interval for the rainfall coefficient contains the truth in fewer than half of outbreaks once the epidemic is a sum of two rather than one — a structural feature these data demonstrably have.

The multiverse interval degrades too, and we report that rather than the headline alone. It falls from 97.5% to 90.0%, so it is no longer calibrated: combining defensible analyses cannot recover from a structural error that all of them share. What it does is absorb most of the damage, turning a 53-point shortfall into a 5-point one. The recommendation survives its hardest test as an improvement and not as a solution, and a reader should carry the 90% rather than the 97.5%.

4.13 What our own recommendation says about these outbreaks

A paper that tells a field its methods are unreliable owes that field its own answer, computed its own way. Otherwise the work is a refusal, and worse, invites the reading that climate does not drive dengue transmission — which is not what any of this shows.

Applying the recommendation of Section 4.8 to the 221 outbreaks — negative binomial only, abstaining where $|\Delta\text{AIC}| < 4$:

Table 20: The verdict this paper would report, under its own rule.

	Outbreaks	Share
Climate forcing supported	118	53.4%
No climate forcing	19	8.6%
Analyses still disagree	49	22.2%
Inconclusive: no analysis speaks	35	15.8%

A clear verdict for 137 of 221 outbreaks, and 86% of those support climate forcing. Where these data can answer the question, the answer is usually yes. The claim of this paper is not that climate forcing is absent; it is that the confidence with which the question is currently answered is not earned, and that in **38% of outbreaks** it cannot be answered at all from a single season of surveillance by this method.

That last figure is the one we would ask a reader to carry into their own work. It is not a failure to report a result. It is the result: a substantial fraction of the outbreaks on which this literature is built do not contain enough information to settle the question they are being asked.

4.14 Checks

Not a small-sample artefact. By quartile of reported cases the unstable share is 94.6%, 90.9%, 89.1% and 92.7%; by quartile of series length, 89.5%, 93.2%, 96.1% and 88.9%. Neither ordering is monotone and no quartile falls below 88%. This is not a problem of thin data.

Not an artefact of our own selection rule. This study applied a peak-prominence threshold when choosing windows and must meet the standard it sets for others. Recomputing the headline on progressively stricter subsets gives 91.9%, 91.8% and 94.1% ($n = 68$). Tightening our selection does not weaken the result.

The climate grid cell matters, and it is measured rather than conceded. A single representative point supplies the climate for a whole country or province, which earlier versions of this study named as a limitation and did not quantify. Forty outbreaks were therefore refitted under all 144 combinations at a second point one degree of latitude away — about 110 km, well inside the area any of these case series aggregates over. Across 5,664 paired comparisons the verdict changes in **12.0%**, with the endorsement probability essentially unmoved ($0.739 \rightarrow 0.737$). That places the climate location between the covariate choices and the structural ones: above the transmission model, below the observation model, and nowhere near the top.

Read it as an upper bound rather than an estimate. The offset is mechanical and respects no terrain: in flat units the two series differ by a few tenths of a degree, which is the intended comparison, but for Costa Rica they differ by 4.3 °C because the shift crosses high ground — a choice between two climates rather than between two defensible stations for one population. The alternative, hand-picking a second city per unit, would replace a reproducible rule with a gazetteer of our own choices, which is not a substitution this study is in a position to make quietly.

Not optimiser noise. The factorial used three multi-starts warm-started from an anchor fit, a compromise made for runtime. If three starts sometimes miss the optimum — and miss it more often for the climate model, which has two extra parameters — then ΔAIC would move

for reasons unrelated to any analysis choice. Seven outbreaks were therefore refitted twice under the 48-combination design in place when the check was run: as run, and with ten cold restarts and no warm start. The thorough setting does find better optima, so the concern is not empty: it improves the climate fit in 6.1% of comparisons and the constant fit in 3.3%, and in one case by 1,656 AIC units. But it changes the verdict in **0.6%** of 329 paired comparisons, against 8.3% and 38.1% for the weakest and strongest analysis factors *in that same 48-combination design* — both bounds move when the factorial grows, so the comparison has to be made within one design — and every one of the seven outbreaks is unstable under both settings. A more thorough optimiser finds more, and what it finds does not move the conclusion. This check is small — seven outbreaks, run alongside the simulation study on the same machine — and is reported as a bound on optimiser noise rather than a precise estimate of it.

5 Discussion

The result is not that climate does not drive dengue transmission. It is that on this evidence, analysed this way, the question is answered by the analyst more often than by the data. Two defensible analyses of the same outbreak, drawn at random, reach opposite conclusions almost a third of the time. The factors that are genuinely about dengue — whether the model represents the mosquito, and what the biological parameters are — matter least of the six.

The single largest lever is the observation model, which is the choice least likely to be discussed in a paper about climate and dengue, because it is not about climate or dengue. Its mechanism is not the one usually given. Assuming equidispersion in data whose dispersion is a median of six times larger — and more than twenty times in a quarter of outbreaks — does not make the fitted model chase noise: paired within outbreak, the climate coefficients are the same size under both likelihoods, and no larger under Poisson than under a negative binomial in more than half of the pairs. What it changes is the exchange rate. The climate model nests the constant one and so always fits slightly better; without a dispersion parameter to absorb the residual, that same small improvement buys thirteen times as much evidence, and it buys it in one direction only.

That this matters in practice, and not only in principle, follows from what the literature reports about its own habits. The systematic review of 99 dengue prediction models finds 18.3% using Poisson regression and 18.3% linear regression, with the negative binomial not appearing as a category of its own [1]; where the two are compared head to head on mosquito-borne outbreak data the negative binomial is the better model [9]. We do not claim this is what most of the field does — more than a third of that sample is the defensible reading — but the choice that raises the probability of endorsing climate forcing from 0.57 to 0.90 is one a large minority makes, and one that mechanistic fits, which the review does not cover, commonly make in the more extreme form of ordinary least squares.

An independent result points the same way from a different direction. Comparing two meteorological stations for dengue forecasting in Thailand, Khamthong and Phramrung report that “forecasting performance is not determined solely by the strength of marginal climate–dengue associations” [8]: the station whose climate correlated better with incidence forecast no better than the one aligned with the population at risk, the two being “statistically equivalent”. That is the same dissociation reported in Section 4.3 from a different lever — in-sample endorsement of climate forcing runs at 68% and out-of-sample at 35% on identical fits. Two studies with no design in common find that in-sample climate–dengue association overstates what the data support.

The remedy is unglamorous and effective, but it is not the obvious one. Reaching for a stricter criterion makes the problem worse: BIC and the likelihood-ratio test are less stable than AIC here, not more, because they relocate the boundary rather than removing it. What works is abstention over a band centred on zero, because in three outbreaks in five all the dissent lies

inside it. In the other two something dissents confidently and no band reaches it — but there the dissenter is nearly alone, which is why the pairwise disagreement falls to 4% while two outbreaks in five still contain a dissenting cell.

Stated as a principle rather than a number: **report a verdict only where the evidence exceeds the variation your own analysis choices induce in it.** Measuring that variation requires refitting under alternative defensible choices, which costs a factorial and is what this study performed. In its absence, abstaining below $|\Delta\text{AIC}| = 4$ reduces pairwise disagreement from 29.8% to 4.0% at no cost in verdicts delivered. This asks nothing new of the field — only that existing guidance on interpreting ΔAIC be applied to a class of question where it currently is not.

What we recommend, concretely. Four things. The first three act on the main effect of the analysis, which is 16% of the variation in the verdict; only the fourth touches the interaction, which is 61%, and only the fourth has been validated against a known truth. That is the order to read them in, and it is not the order in which we thought of them.

1. **Use an overdispersed observation model.** This is worth 45 points of separation between a real effect and none, and no other choice examined here is worth a tenth of that. A Poisson likelihood on weekly dengue counts endorses climate forcing on six datasets in seven that contain no climate signal at all. If a negative binomial is not used, the reason should be stated.
2. **Report no verdict from the sign of ΔAIC .** Abstain within ± 4 . It costs nothing in outbreaks answered and cuts the chance that two analyses disagree from 29.8% to 4.0%. Do not substitute a stricter criterion: BIC and the likelihood-ratio test are both worse here, because they relocate the boundary instead of widening it.
3. **State the choices that were made.** The observation model, the functional form of each covariate response, every lag, and how much of the series was fitted, the compartmental structure, and the values at which the biological parameters were fixed. Between the most and least credulous combination of these lies 56 percentage points of endorsement rate. A reader cannot place a published result within that range unless the paper says where it sits.
4. **Refit under a handful of alternatives and report an interval that carries the result.** Not the whole factorial: **eight analyses suffice**, and four already reach nominal coverage. Combining them by Rubin’s rules — within-analysis variance plus between-analysis variance, the multiple-imputation decomposition — turns a 95% interval that covers 71% of the time into one that covers 97%, at 1.5 to 2.2 times the width. Eight fits of each model is a few minutes on one processor. This is the only item here validated against a known truth rather than argued for, and it is the one we would keep if only one survived. Which eight matters, but not as one would guess: vary the choices that construct the covariate you are reporting on. Fixing the observation model costs the interval nothing; fixing the rainfall lag costs the rainfall coefficient nine points of coverage. Two independent results converge on the lag: it is the choice whose effect is least predictable from anything visible before fitting (0.1% of its variation), and the one that cannot be held fixed without losing coverage. Neither was designed to test the other.

None of this is new statistical advice. All of it is standard, and the contribution here is the measured cost of not following it in one specific and consequential class of question.

Limitations. Reported cases are not infections, and reporting effort varies within and between the surveillance systems compiled here. The sample is not representative of dengue globally: 34 of the 88 countries in the weekly subset contribute a usable window. Which requirement excludes the rest is worth stating exactly, because an earlier version of this paragraph guessed and guessed wrong. Of the 54 excluded countries, **17** have no gap-free run of 30 weeks — the mechanism the paragraph named — while **37** report continuously enough and are excluded because no wave in their record is large or sharply peaked enough to fit. The dominant selection is on *outbreak size and shape*, not on reporting continuity, and it points the opposite way from the concern as written: large single-wave epidemics are the ones these models handle best, so the instability measured here is what survives on the most favourable data available rather than on the hardest. The retained countries’ longest gap-free run is nevertheless about seven times the excluded countries’, so selection on surveillance continuity is real and second. Whether either group is poorer is not something these data record, and we no longer say it. A single climate location represents each reporting unit; in the Pakistani case study that choice moved R_0 by more than the sampling uncertainty, though the evidence that it changes predictive performance is weaker [8]; Section 4.14 measures how often it changes a verdict, and finds 12%. The factorial covers six choices and is not exhaustive: the set of climate covariates, the window-selection rule and the criterion itself are further degrees of freedom. Each factor was also given only two or three defensible settings, not the full range a literature contains, so every figure here is a lower bound on the instability available to an analyst. Finally, all fits use a single-strain model for a four-serotype disease; representing serotype-specific immunity would fork the multiverse again, and on single-wave data would likely do so without being identifiable.

Data and code availability

All analyses are reproducible from <https://github.com/amnadilber/dengue-multiverse>: 38 numbered pipeline scripts, 266 tests, and a dated analysis log recording every formulation that failed and why, including errors in earlier versions of this analysis that produced plausible wrong numbers rather than failing, and which the log records rather than conceals. Tests check the paper’s quoted numbers against the stored result tables, so a manuscript that has drifted from the pipeline fails the suite. A separate step checks the claims that are not numbers, because five interpretive sentences in earlier drafts were wrong while every figure around them was right. Case data are from OpenDengue v1.3 (CC-BY 4.0); climate data from NASA POWER (public domain). Raw data are not redistributed; download scripts record source URLs and checksums.

Appendix: what can be identified at all, a worked case

The global study asks whether a conclusion is stable. It does not ask whether the underlying quantities are identifiable, which we examined in detail on Pakistani surveillance — a national series covering the complete 2013 wave and provincial series from 2021.

Transmission is identified; catchment size often is not. Profile likelihoods [21] bound R_0 and the population at risk in the longer windows, but in a nineteen-week provincial window the population profile under the climate model is flat across a twelvefold range, while the constant model determines it to within a factor of two. Adding climate parameters to a short series destroys the identifiability of a parameter the simpler model recovers.

Spatial aggregation depresses R_0 . Fitting districts separately and then fitting their sum gives different answers: a bias of -2.2% in one province and -28.8% in another, where the aggregate estimate of 1.14 falls below both constituent districts, 1.25 and 1.96. Component epidemics peak at different times, and a homogeneously mixing model can only read the broader sum as slower transmission.

An independent check confirms it. Bounding R_0 from the observed early growth rate — using no compartmental fit, and reporting both the fixed and exponentially distributed generation-interval bounds — gives 2.91–6.77 nationally against a fitted 1.37. Every fitted value falls below the corresponding lower bound. A homogeneous model ties growth rate to final size through a single R_0 ; these epidemics rise faster than their eventual size permits, which is the same spatial heterogeneity seen from a second direction.

And the explanation was tested, not asserted. Spatial heterogeneity is the natural account of both observations, but naming a mechanism is not demonstrating it. A two-patch model was therefore fitted to the Khyber Pakhtunkhwa *aggregate* alone — given no information about which districts contributed, how many there were, or when each peaked. It beats the single-patch fit by $\Delta\text{AIC} = -8.6$, recovers transmission rates of 1.72 and 2.53 that bracket the two districts instead of falling below both, clears the growth-rate lower bound the single-patch fit violated, and infers a 46-day offset between its components. Three predictions of the heterogeneity account, each checkable and each met, from a model that was told only the sum.

Fitted transmission rates from aggregated surveillance should therefore be read as lower bounds, not as characteristic values.

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